3. Greatest Common Divisor (GCD)

Problem Statement

Write recursive code to calculate the Greatest Common Divisor (GCD) of Two Positive Numbers.

The greatest common divisor (GCD) of two positive integers A and B is the largest positive integer that divides both A and B without leaving a remainder.

Let's see some example inputs/outputs for this example:

Input(s)	Output(s)	Explanation
A = 12, B = 18	GCD = 6	The factors of 12 are [1, 2, 3, 4, 6, 12], and the factors of 18 are [1, 2, 3, 6, 9, 18]. The common factors between 12 and 18 are [1, 2, 3, 6], and the largest common factor is 6.
A = 25, B = 15	GCD = 5	The factors of 25 are [1, 5, 25], and the factors of 15 are [1, 3, 5, 15]. The common factors between 25 and 15 are [1, 5], and the largest common factor is 5.
A = 40, B = 60	GCD = 20	The factors of 40 are [1, 2, 4, 5, 8, 10, 20, 40], and the factors of 60 are [1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60]. The common factors between 40 and 60 are [1, 2, 4, 5, 10, 20], and the largest common factor is 20.

Try it yourself

Try solving this question here:

Code

Here is the code for this algorithm:

Java

```java
public class Solution {
  public static int calculateGCD(int A, int B) {
    // TODO: Write your code here
    return 0;
  }
}
```

```
7
```

Test Case | Run Code Result | Stdout

```
12
18
```

Use Example Testcases | Run | Submit

← Previous

Solution: Factorial

Next →

Solution: Greatest Common Divisor (GCD)

Mark as Completed